Damian Ramirez

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Education

The University of Texas Rio Grande Valley Expected December 2026
Bachelor of Science in Computer Science *GPA: 3.63*

Professional Experience

Texas Manufacturing Assistance Center September 2025 – Present
Software Engineering Intern

- Engineered a manufacturing assessment platform using Python, FastAPI, and PostgreSQL, reducing assessment-to-report turnaround from approximately 3 days to 3 hours during internal testing.
- Developed a 60-question assessment engine across 10 operational categories, automating performance scoring, report generation, and historical tracking for repeat client evaluations.
- Implemented REST APIs, SQLAlchemy ORM models, and JWT authentication to securely manage client profiles, assessment records, and evaluation history.
- Automated 5 decision-support outputs, including performance dashboards, Pareto gap analysis, AI-assisted proposals, and 30–60–90-day improvement plans for manufacturing advisors.

Brownsville Public Utilities Board May 2026 – August 2026
Operations Analyst Intern

- Integrated 15+ Excel files, PDF reports, and Oracle data sources into a centralized Power BI application tracking 16 enterprise KPIs across electric, water, and wastewater operations.
- Automated data transformations, KPI calculations, and recurring reporting workflows using Power Query and DAX, replacing decentralized spreadsheets and manual monthly reporting processes.
- Configured automated daily data refreshes, reducing reporting latency from up to 30 days to approximately 24 hours and improving visibility into utility performance.
- Developed 4 real-time Power BI dashboards spanning 15 reporting pages and 16+ interactive visualizations, providing executives enterprise-wide visibility and department managers role-restricted KPI access.

Projects

GridPulse – ERCOT Grid Analytics Platform September 2026 – Present

- Built a Python data ingestion pipeline using the EIA API to retrieve ERCOT electricity demand and forecast data, collecting 30,408 raw records and transforming them into 15,264 hourly observations.
- Implemented PostgreSQL storage and SQL analytics for electricity time-series data, with separate raw and processed data layers for structured ingestion.
- Developed XGBoost time-series forecasting experiments using lag-based features and chronological validation, benchmarking predictions against ERCOT forecasts and a 24-hour persistence baseline.

SBA Loan Default Risk Predictor April 2026

- Developed an XGBoost machine learning model to predict loan default risk across approximately 8,000 Texas SBA loans, achieving 94% accuracy and 0.964 ROC-AUC.
- Integrated SBA loan records with 3 U.S. Census datasets, engineered 608 predictive features, and deployed an interactive Streamlit application with SHAP-based model explanations to interpret loan default risk predictions.

Technical Skills

Programming Languages: Python, SQL, TypeScript, C++
Reporting & Analytics: Power BI, Microsoft Excel, Power Query, DAX, Dashboard Development, KPI Reporting
Databases & Data Processing: PostgreSQL, Oracle, SQLAlchemy, Pandas, NumPy, Data Integration
Software Development: React, Next.js, FastAPI, REST APIs, Git, GitHub
Machine Learning: Scikit-learn, XGBoost, SHAP, Streamlit